

ACN Practical 1

1. What information can you typically obtain from a WHOIS query for an IP address?

Ans. A WHOIS query for an IP address can typically provide information about who controls or is responsible for that IP address and the network it belongs to.

Common information includes:

1. IP Address
2. Organization/ISP
3. Network Range
4. ASN (Autonomous System Number)
5. Country/Region
6. Registration Information
7. Abuse Contact
8. Allocation Dates
9. WHOIS/RIR Server

2. Which subnet mask corresponds to a network with 256 possible IP addresses?

Ans. A network with 256 possible IP addresses requires 8 host bits, because:

$$2^8 = 256$$

Therefore, the subnet mask is : 255.255.255.0 (/24)

3. List the key roles of ICANN.

Ans. The key roles of ICANN (Internet Corporation for Assigned Names and Numbers) are:

1. Domain Name Management
2. IP Address Allocation
3. Root Server System Coordination
4. Domain Name Registries and Registrars
5. Protocol Parameter Coordination
6. Internet Stability and Security

4. State the Subnet Address. What is the purpose of assigning a subnet mask to a network device?

Ans. A subnet address is the network address that identifies a specific subnet within a larger IP network. It is obtained by performing a bitwise AND operation between the IP address and the subnet mask.

Purpose of assigning a subnet mask:

1. Identifies the network and host portions of an IP address.
2. Determines the subnet to which a device belongs.
3. Allows devices to determine whether another device is on the same network or a different network.

4. Helps divide a large network into smaller subnetworks, improving network organization and management.
5. Reduces unnecessary network traffic by limiting broadcast domains.

5. If a network device has the IP address 10.10.20.45 with subnet mask 255.255.255.0, what is the network address?

Ans. Given:

- IP Address: 10.10.20.45
- Subnet Mask: 255.255.255.0 (/24)

The first 24 bits represent the network, while the last 8 bits represent the host.

Therefore:

Network Address = 10.10.20.0

6. If a network device has the IP address 172.16.25.100 with subnet mask 255.255.255.0, what is the network address?

Ans. Given:

- IP Address: 172.16.25.100
- Subnet Mask: 255.255.255.0 (/24)

The first three octets represent the network portion, and the last octet represents the host portion.

Therefore:

Network Address = 172.16.25.0

7. State the concept of fragmentation in IPv4.

Ans. IPv4 fragmentation is the process of dividing a large IP packet into smaller fragments when the packet size is greater than the Maximum Transmission Unit (MTU) of the network through which it must travel.

Each fragment is transmitted separately and contains information that allows the destination device to reassemble the original packet.

Example:

If a 4000-byte IPv4 packet must pass through a network with an MTU of 1500 bytes, it may be divided into multiple smaller fragments.

8. For the given IP address: 193.168.10.1

find:-

i) The class of IP address

ii) Identify Network address

iii) Identify Host address

iv) Identify First address.

v) Identify Last address

vi) Find subnet address.

Ans. i) Class of IP Address

The first octet is 193.

Therefore, 193.168.10.1 is a Class C IP address.

ii) Network Address

For a Class C address, the default subnet mask is 255.255.255.0 (/24).

Network Address = 193.168.10.0

iii) Host Address

The host portion is the last octet.

Given IP: 193.168.10.1

Host Address = 0.0.0.1

iv) First Address

The first usable host address is:

193.168.10.1

v) Last Address

The broadcast address is 193.168.10.255, so the last usable host address is:

193.168.10.254

vi) Subnet Address

Using the default Class C subnet mask 255.255.255.0:

Subnet Address = 193.168.10.0